

domains. ED, PID, UFD.



Def 3. Let R be a commutative ring that is a domain. A function $N: R \rightarrow \mathbb{N}$ is called a norm if $N(0) = 0$.

Def 4. Let R be a commutative ring that is a domain. Then R is an Euclidean domain if there is a norm $N: R \rightarrow \mathbb{N}$ such that for all $a, b \in R$, $b \neq 0$.

$$a = qb + r \quad \text{for} \quad r = 0 \quad \text{or} \quad N(r) < N(b)$$

eg. $\mathbb{Z}$ is an Euclidean domain

with norm being the absolute value.

Lemma 10. Let R be a UFD.

Then $r \in R$ is prime iff it is irreducible

proof: It suffices to prove irreducibles

are primes. Let $r \in R$ be irreducible

and assume $r \mid ab$, or $rn = ab, n \in R$

Since R is UFD. we can factorize

$$a = p_1 p_2 \cdots p_s, \quad s \in \mathbb{N}$$

$$b = q_1 q_2 \cdots q_t, \quad t \in \mathbb{N}$$

$$n = m_1 m_2 \cdots m_k, \quad k \in \mathbb{N}$$

where p_i, q_j, m_k are all irreducibles

$$\text{So } r m_1 m_2 \cdots m_k = p_1 \cdots p_s q_1 \cdots q_t$$

Since R is UFD,

r is an associate of some p_i

or some q_j , which implies

$$r \mid a \quad \text{or} \quad r \mid b.$$

Theorem 11. Every PID is UFD.

proof: We first show every nonzero

nonunit element $r \in R$ can be written

as a product of finitely many irreducibles

If r is irreducible then done.

if not, write $r = r_1 r_2$ where r_1, r_2 nonunit

Repeat the process to r_1 and r_2

Say if r_1 reducible, then $r_1 = r_{11} r_{12}$

and we stop when there all irreducibles

We have to show such process stops

after finitely many steps. Note we

have sequence of ideals

$$(r) \subset (r_1) \subset (r_{11}) \subset \dots$$

It suffices to show in PID,

